

Shubh Gupta

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Artificial Intelligence

CGPA: 8.73

October 2022 – May 2027

INTERNSHIP EXPERIENCE

AI Intern – Information Technology Department

MP Birla Group (Birla Corporation Limited)

Satna, Madhya Pradesh

October 2025 – Dec 2025

- Built a Python + OpenCV + Tesseract system to automate form digitization, syncing extracted data to Google Sheets and reducing manual data entry effort by **85%**.
- Developed a Streamlit-based review tool for correcting OCR errors, ensuring data accuracy before final submission.
- Achieved **92% extraction accuracy** on filled forms through image preprocessing pipelines and iterative model refinement.

PROJECTS

Verba | Django, Pinecone, Gemini 2.0, Groq LLaMA 3, Supabase | [GitHub](#) | [Live Demo](#)

Nov 2025

- Built a production-ready RAG pipeline supporting PDF, DOCX, XLSX, CSV, and scanned images — using Gemini 2.0 Flash Vision OCR, Google Generative AI embeddings, and Pinecone serverless vector storage for semantic retrieval.
- Engineered a parallel OCR pipeline using `ThreadPoolExecutor` achieving **5x faster** scanned PDF processing, with async background ingestion to eliminate server timeouts and seamless UI auto-polling.
- Integrated Supabase for PostgreSQL storage, S3 file storage, and Google OAuth authentication; deployed on Render with Django 5 + Gunicorn and smart rate limiting (50 messages per 3-hour rolling window).

Deep Image Steganography | Python, TensorFlow, PyTorch, OpenCV | [GitHub](#) | [Live Demo](#)

April 2025

- Developed a generative AI-based image steganography system using GAN, VAE, and DDPM models to securely hide images within cover images with minimal visual distortion.
- Built encoder-decoder and diffusion pipelines for high-fidelity steganographic embedding and extraction of hidden data.
- Improved model robustness by **18%** using adversarial loss tuning and diffusion-based denoising for secure covert communication.

FaceGuard AI | FastAPI, YOLOv8, FaceNet, React.js, WebSockets | [GitHub](#)

Sep 2025

- Built a full-stack real-time face recognition system with live video streaming using FastAPI backend and React.js frontend connected via WebSockets.
- Implemented YOLOv8 for face detection and FaceNet embeddings with one-shot learning for accurate user identification from minimal training samples.
- Optimized face-matching latency by **40%** using FAISS-based vector search and asynchronous inference for real-time performance.

TECHNICAL SKILLS

Programming: Python, C++, SQL

Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Operating System

AI & Machine Learning: Deep Learning, GANs, NLP, Computer Vision, Generative AI, RAG

Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, FastAPI, Django

Tools & Platforms: Git, GitHub, Docker, Pinecone, Supabase, Tesseract OCR, Google Colab, VS Code

ACHIEVEMENTS

- **NASSCOM AI Hackathon:** Semi-finalist
- **TATA Imagination Challenge 2025:** Finalist